

RATHISRI R

Computer Science Engineer | AI & ML

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PROFILE SUMMARY

B.Tech CSE (Hons.) student specializing in AI & ML, GPA 9.52, with hands-on experience in Python, Django, React, and IoT/embedded systems. Published researcher (ISBN-certified) skilled in building end-to-end AI, full-stack, and IoT solutions.

EDUCATION

B.Tech in Computer Science and Engineering (Hons.) — AI & ML Specialization

Periyar Maniammai Institute of Science and Technology, Thanjavur | 2023–2027 | GPA: 9.52

TECHNICAL SKILLS

- **Languages:** C, Python, Java (OOP)
- **Web Development:** HTML, CSS, JavaScript, React, Django (REST APIs, WebSockets, JWT Auth)
- **IoT / Hardware:** ESP32, ESP32-CAM, Arduino, LiDAR, Ultrasonic & DHT Sensors, Servo Control
- **CS Fundamentals:** Data Structures & Algorithms, OS, Computer Networks, DBMS
- **Tools:** Git/GitHub, Linux, ThingSpeak, MATLAB, VS Code

PROJECTS

AI-Based Cotton Harvesting Machine

- Engineered an ESP32-CAM AI vision system to detect and classify ripe cotton bolls, integrated with a robotic arm, vacuum suction, and LiDAR/ultrasonic obstacle avoidance.

AI Chatbot Web Application

- Built a full-stack chatbot using Django and JavaScript with Gemini/OpenAI API integration, JWT authentication, an admin dashboard, and real-time chat via WebSockets.

Neuro Cardiac Arrest Monitoring System

- Developed an IoT-based system tracking heart rate and neurological signals with automated real-time alerts for critical anomalies; published as an ISBN-certified paper.

Smart Parking System (IoT)

- Built an Arduino-based real-time parking slot detection system with servo gate control, LCD status display, and ThingSpeak cloud monitoring.

LFU Cache Implementation

- Implemented an LFU cache in Python using a HashMap and Doubly Linked List, achieving $O(1)$ time complexity for get/put operations.

CERTIFICATIONS & ACHIEVEMENTS

- TANCAM Women Hackathon — Selected among Top 50 teams out of 1000+ teams
- Robotics and IoT Course — Arduino, ESP32 & embedded systems design
- AI Chatbot Development — Vdart Technologies Pvt Ltd, Trichy
- Web Development (Django, HTML, JS) • Data Structures Using C
- Typewriting (English & Tamil) — Junior & Senior, Distinction
- Hindi Language — Completed All 8 Levels

PUBLICATION

- Neuro Cardiac Arrest Monitoring System — ISBN Certified Publication

ADDITIONAL INFORMATION

- **Languages:** Tamil (Native), English (Proficient)
- **Soft Skills:** Problem-Solving, Analytical Thinking, Collaboration, Communication, Adaptability